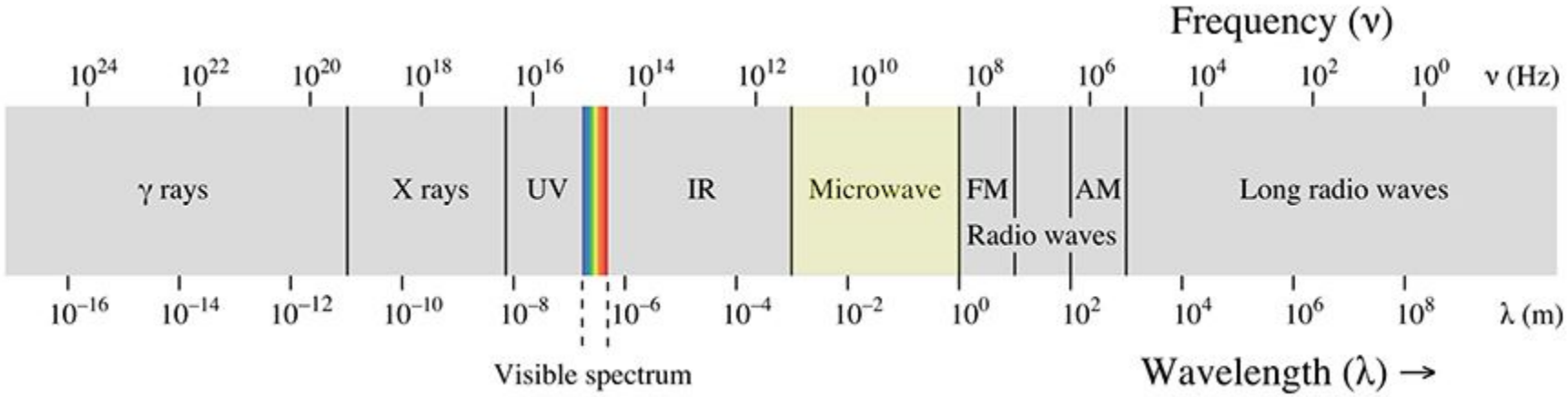
Intro to SAR

Differences between SAR and optical

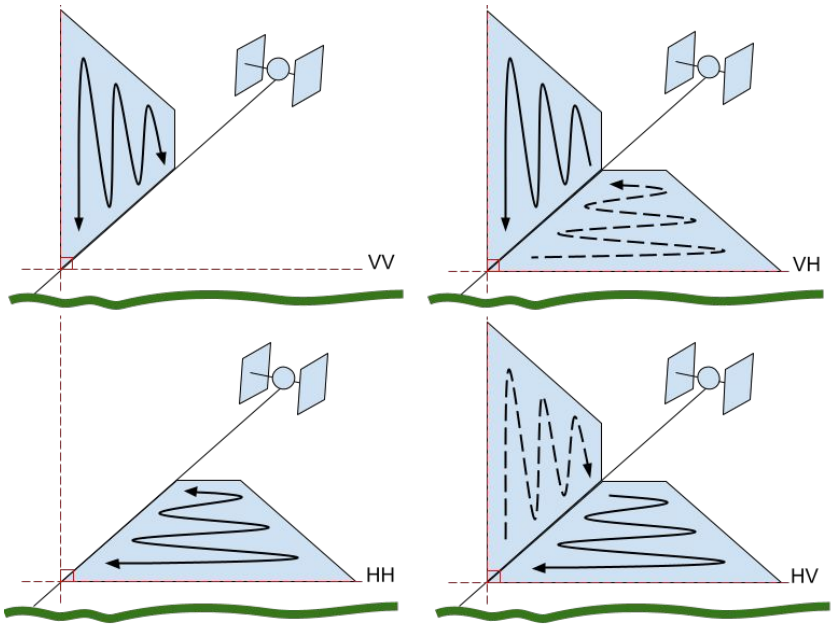
- Optical: reflected solar radiation, intuitive appearance, limited by cloud/daylight.
- SAR: active microwave sensor, day/night, (effectively) all weather, sensitive to moisture and structure.
- SAR does not look like a photograph; it measures backscatter.
    a. Measurement vs picture - amplitude and phase are measured



SAR transmits an encoded burst, called a chirp, of electromagnetic energy (Figure 1) and then listens for the return signal, called echoes. The wavelength of this chirp is in the centimeter range (Microwave region), with X-band (~3 cm), C-band (~6 cm), and L-band (~23 cm) all in common use. (Credit ASF https://hyp3-docs.asf.alaska.edu/guides/introduction_to_sar/)

What SAR actually measures

- Sensor sends a pulse and records the echo.
- SAR records amplitude and phase.
- For this lab, we only care about amplitude/backscatter intensity, not phase.
  a. Copernicus S1_GRD is pre-processed and has phase removed.
- Backscatter depends on roughness, structure, moisture, wavelength, polarization, and geometry.



SAR can transmit (first letter) and receive (second letter) in Vertical or horizontal polarizations, but some radars can only transmit/recieve in one (single-pol), and others can transmit/recieve in both (dual-pol).

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Side-looking geometry: why SAR images look strange

- SAR is side-looking, not nadir (straight down) looking.
- Range, azimuth, slant range, look angle/incidence angle.
- This is the single most important geometric idea.

Slant Range Distance

- The direct distance between the sensor and the target

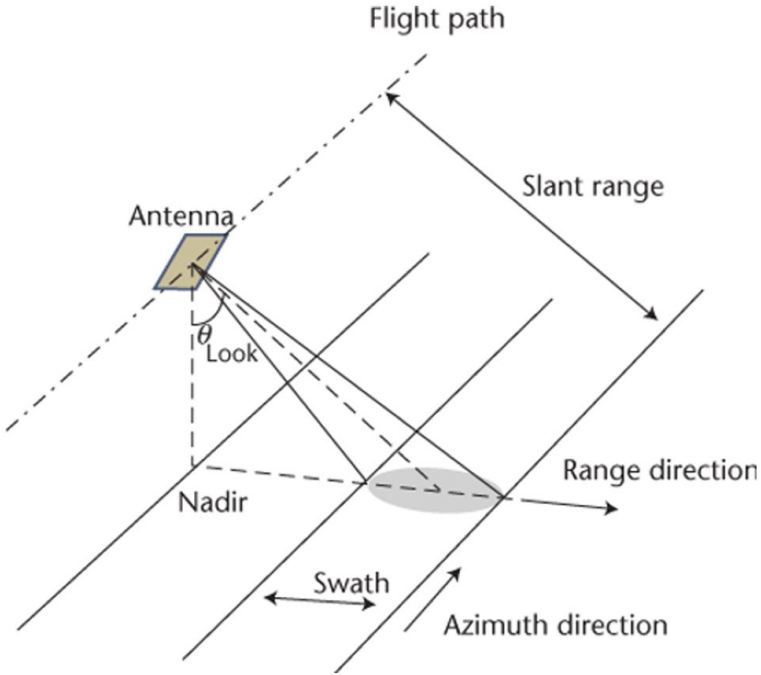
Range Direction (Across Track)

- The direction perpendicular to the motion of the radar sensor

Azimuth Direction (Along Track)

- The direction parallel to the motion of the radar platform as it moves over the target area

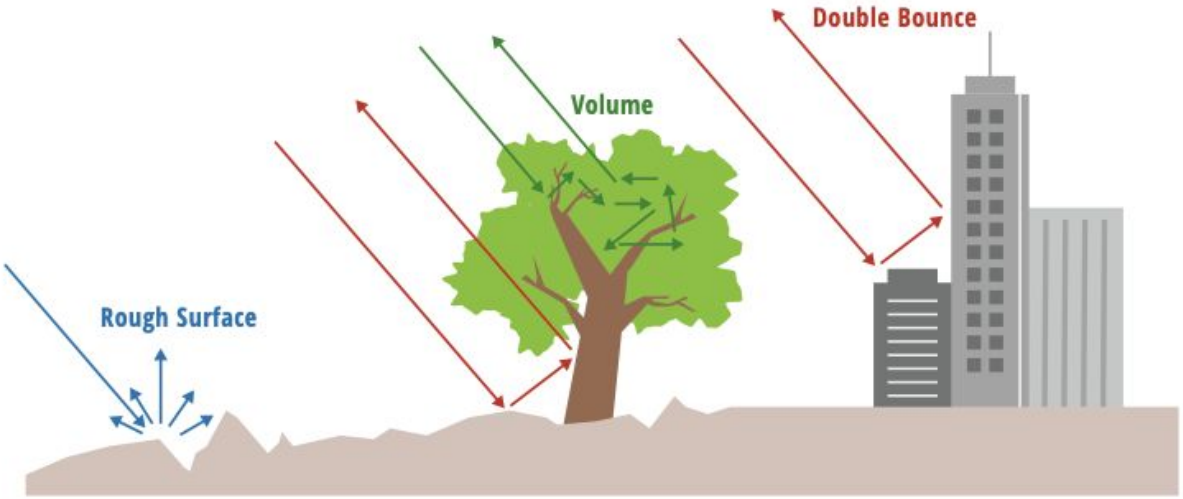
https://appliedsciences.nasa.gov/sites/default/files/2024-11/SAR_2024_Part1_English_FINAL.pdf



Why things are bright or dark: backscatter mechanisms

Why things are bright or dark: backscatter mechanisms

- Smooth water / smooth surfaces: specular reflection → usually dark.
- Rough surfaces: more diffuse return → brighter.
- Vegetation: volume scattering → stronger VH return.
- Urban corners/buildings: double bounce → often very bright.



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How to read a SAR image fast

- Water usually dark.
- Urban often bright.
- Forest/vegetation often intermediate and texture-rich.
- SAR brightness is not “albedo”; it is interaction between radar and target geometry/materials.

What data we are using in the lab

- Sentinel-1 GRD
- VV and VH
- GRD is amplitude/intensity-style imagery, not the complex SLC product (no phase).
- Today we are doing mapping / interpretation / simple change, not interferometry.
- We use Google Earth Engine, which has a collection of pre-processed GRD images